Customer Behavior Analysis Report

1. Business Context

The company is a retail business operating across multiple product categories (Clothing, Accessories, Footwear, Outerwear).
Management wants to better understand **customer purchasing behavior** to (improve revenue, increase customer satisfaction, Strengthen long-term customer loyalty)

The data was previously under-utilized, and decisions were made without a clear, data-driven view of customer behavior.

2. Objective of the Analysis

The objective of this project is to answer key business questions such as:

- Who generates the most revenue?
- Which products and categories perform best?
- How discounts, shipping, and subscriptions affect spending
- How customer behavior differs by age group and gender

The goal is to translate data into **clear business insights and actionable recommendations**.

3. Data & Tools Used

Data Source:

Using transactional data from 3900 purchases across various product categories:

- Rows: 3900;
- Columns: 18;
- Key Features:
  - Customer demographics (Age, gender, Location, Subscription Status)
  - Purchase details (Item purchase, Category, Purchase Amount, Season, Size,Color)
  - Shopping behavior(Discount Applied, Promo Code Used, Previous Purchases, Frequency of Purchases, Review Rating, Shipping type)
  - Missing Data: 37 values in review rating column

Tools

- ✓ **Python**: data cleaning, validation, transformation
- ✓ **MySQL**: business analysis and answering stakeholder questions
- ✓ **Power BI**: interactive dashboard and visualization

4. Exploratory Data Analysis using Python

We began with data preparation and cleaning in python pandas and numpy:

Data loading: imported dataset using pandas.

Initial Exploration: used df.info() to check structure and df.describe() for summary statistics

[67]:

	Customer ID	Age	Purchase Amount (USD)	Review Rating	Previous Purchases
count	3900.000000	3900.000000	3900.000000	3863.000000	3900.000000
mean	1950.500000	44.068462	59.764359	3.750065	25.351538
std	1125.977353	15.207589	23.685392	0.716983	14.447125
min	1.000000	18.000000	20.000000	2.500000	1.000000
25%	975.750000	31.000000	39.000000	3.100000	13.000000
50%	1950.500000	44.000000	60.000000	3.800000	25.000000
75%	2925.250000	57.000000	81.000000	4.400000	38.000000
max	3900.000000	70.000000	100.000000	5.000000	50.000000

Missing Data Handling: checking for null values and imputed missing values in the Review rating column using the median rating of each product category.

Column Standardization: renamed columns to **snake case** for better readability and documentation.

Feature Engineering:

- Created **age_group** column by binning customer ages
- Created **purchase_frequency_days** column from purchase data

Data Consistency check: verified if discount_applied and promo_code_used were redundant; dropped promo_code_used

Database Integration: Conneted Python script to MySQL and loaded the cleaned DataFrame into the database for SQL analysis.

5. Data Analysis using SQL (Business Transaction)

We performed structured analysis in MySQL to answer eke business question

1. ***Revenue by gender:*** Compared total revenue generated by male vs female customers

Result Grid | Filter Rows:

gender	Revenue
Male	157890
Female	75191

2. ***High Spending Discount*** Users: Identified customers who used discounts but still spent above the average purchase amount.

customer_id	purchase_amount
2	64
3	73
4	90
7	85
9	97
12	68
13	72
16	81
20	90
22	62
24	88
29	94
32	79

3. ***Top 5 Products Rating:*** Found products with the highest average review ratings

item_purchased	avg_review_rating
Gloves	3.86
Sandals	3.84
Boots	3.82
Hat	3.8
Handbag	3.78

4. ***Shipping Type Comparison:*** Compare the average Purchase Amounts between Standard and Express Shipping.

shipping_type	avg_purchase_amount
Express	60.48
Standard	58.46

5. ***Subscribers Vs Non-subscribers:*** Compare average spend and total revenue between subscribers and non-subscribers.

subscription_status	avg_spend	total_revenue
Yes	59.49	62645
No	59.87	170436

6. ***Discount-Dependent Products:*** Identified 5 products with the highest percentage of discount purchases

item_purchased	discount_prct
Hat	50.00
Sneakers	49.66
Coat	49.07
Sweater	48.17
Pants	47.37

7. ***Customers Segmentation*****:** Classified customers into New, Returning, and Loyal based on purchases history.

Customer_Segment	Number_Of_Customer
New	83
Returning	701
Loyal	3116

8. ***Top 3 Products per Category:*** Listed the most purchased products within each category.

category	item_purchased	rating_rank
Accessories	Jewelry	1
Accessories	Sunglasses	2
Accessories	Belt	3
Clothing	Blouse	1
Clothing	Pants	2
Clothing	Shirt	3
Footwear	Sandals	1
Footwear	Shoes	2
Footwear	Sneakers	3
Outerwear	Jacket	1
Outerwear	Coat	2

9. ***Repeat Buyers & Subscription:*** Checked whether customers with >5 purchases are likely to subscribe.

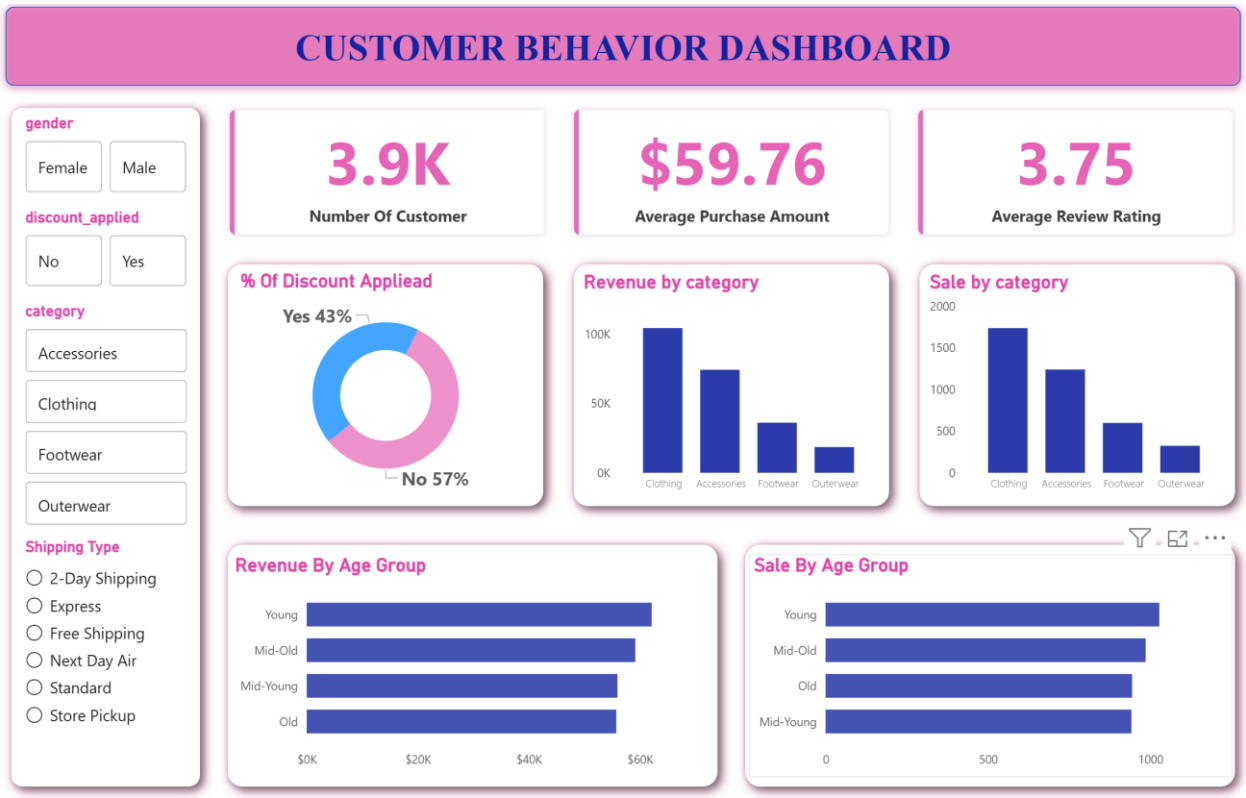
subscription_status	count
Yes	958
No	2518

10. ***Revenue by Age Group:*** Calculate total revenue contribution of each age group.

age_group	revenue
Young	62143
Mid-Old	59197
Mid-Young	55978
Old	55763

6. Dashboard in Power Bi

Finally, we built an interactive dashboard in Power Bi to present insight visually.



7. Business Recommendations

1. Focus marketing efforts on Clothing and Accessories, as they drive the majority of revenue.
2. Strengthen subscription programs to convert repeat buyers into loyal customers.
3. Use targeted discounts rather than blanket promotions to protect margins.
4. Improve customer experience (delivery, service) to raise review ratings.
5. Segment customers by age and behavior to personalize offers and communication.

8. Conclusion

This analysis demonstrates how customer behavior data can be used to make informed business decisions.
By combining **Python for data preparation**, **SQL for analysis**, and **Power BI for visualization**, the company gains a clear view of what drives revenue and customer satisfaction. The insights from this project can directly support management decisions related to pricing, marketing strategy, and customer retention.